

RONSHEE CHAWLA

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EDUCATION

- **University of Texas at Austin (UT)**
PhD in Electrical and Computer Engineering August 2017 – Present
MS in Electrical and Computer Engineering August 2017 – May 2019
Advisor: Prof. *Sanjay Shakkottai*
Current GPA: 4.0/4.0
- **California Institute of Technology (Caltech)** September 2016 – August 2017
Graduate Student in Electrical Engineering
GPA: 3.7/4.0
- **Indian Institute of Technology (IIT) Indore, India** July 2011 – May 2015
B. Tech. in Electrical Engineering
Advisor: Prof. *Prabhat K. Upadhyay*
GPA: 9.89/10.0

RESEARCH INTERESTS

- Design and analysis of algorithms for decentralized/distributed machine learning
- Reinforcement learning theory and algorithms
- Bandit algorithms

PUBLICATIONS (* denotes equal contribution)

- *Collaborative Multi-Agent Heterogeneous Multi-Armed Bandits*
R. Chawla, Daniel Vial, Sanjay Shakkottai, R. Srikant
International Conference on Machine Learning (ICML) 2023 (28% accepted)
- *Multi-Agent Low-Dimensional Linear Bandits*
R. Chawla, Abishek Sankararaman, Sanjay Shakkottai
IEEE Transactions on Automatic Control, 2023
- *The Gossiping Insert-Eliminate Algorithm for Multi-Agent Bandits*
R. Chawla*, Abishek Sankararaman*, Ayalvadi Ganesh and Sanjay Shakkottai
International Conference on Artificial Intelligence and Statistics (AISTATS) 2020 (30% accepted)
- *Outage Performance and Location Optimization for Traffic-Aware Two-Way Relaying with Direct Link*
Suneel Yadav, R. Chawla and Prabhat K. Upadhyay
International Conference on Signal Processing and Communications (SPCOM) 2016
- *Outage Analysis of Cellular Two-Way Relaying with Spectrum Sharing in Nakagami- m Fading*
R. Chawla, M. Mounika Reddy and Prabhat K. Upadhyay
Proceedings of 18th International Symposium on Wireless Personal Multimedia Communications (WPMC) 2015

TECHNICAL SKILLS

- **Programming languages:** C++, Python
- **Machine learning frameworks:** PyTorch
- **Softwares:** MATLAB

WORK EXPERIENCE

- **PhD Research Software Engineer Intern** Summer 2022
Uber Technologies Inc., Sunnyvale, CA USA
Worked in Marketplace Investments Modeling team, where I proposed a new deep neural network model for optimal allocation of budget across driver incentives to maximize the revenue, in a computationally efficient manner
- **Scientist-B** August 2015 – May 2016
Aeronautical Development Establishment (ADE), DRDO, Bengaluru, India
Flight Test Telecommand and Tracking Division
- **Internship** May 2014 – July 2014
RWTH Aachen University, Germany
Worked on automatic extraction of layer-specific data-flow configurations for automatic scheduling and mapping of kernels for the Layers architecture

TEACHING EXPERIENCE

- *Probability, Statistics and Random Processes*, Teaching Assistant, Fall 2017
Online Learning (Graduate), Teaching Assistant, Fall 2019
– Duties included conducting office hours, and setting homework and exam problems

TALKS

- *The Gossiping Insert-Eliminate Algorithm for Multi-Agent Bandits*
Invited talk at Vrbo, Austin, TX February 2020

ACADEMIC SERVICE

- Reviewer for IEEE Conference on Decision and Control 2022, 2023
- Reviewer for IEEE Control Systems Letters 2022
- Reviewer for Conference on Neural Information Processing Systems (NeurIPS) 2022
- Reviewer for International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

HONORS AND ACHIEVEMENTS

- First rank in Electrical Engineering PhD Qualifying Exam at Caltech, January 2017.
- **President of India Gold Medal**, first rank in IIT Indore, class of 2015.
- DAAD WISE scholarship for summer internship at RWTH Aachen University, Germany, May 2014 – July 2014.
- Offered summer fellowship by Indian Academy of Sciences, May 2014 – July 2014 (declined).
- Academic Excellence Awards, IIT Indore, 2012 – 13, 2013 – 14 and 2014 – 15.

GRADUATE COURSEWORK

- **Machine Learning:** Learning Systems, Special Topics in Unsupervised Learning (GANs), Statistical Learning Theory, Fair Transparent Machine Learning
- **Mathematics:** Probability and Stochastic Processes, Advanced Probability, Convex Optimization (Theory and Algorithms), Information Theory, Applied Linear Algebra, Random Matrices, Coding Theory, Mathematics of Signal Processing, Queueing Theory